llmXive Automated Review of ArcANE: Do Role-Playing Language Agents Stay in Character at the Right Time?

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong claims regarding the validation of its dataset and the sensitivity of its metrics, but the supporting evidence in the text and tables occasionally overgeneralizes or lacks precise alignment.

First, in Section 3.1 ("Character Arc Construction"), the authors claim a rigorous validation process where "three human annotators independently assess axes; only those with a 2-of-3 majority enter the set." Later, in Section 5.2 ("Evaluation validation"), the paper reports "Human plausibility is 87.1%." While these numbers are likely related, the text does not explicitly clarify if the 87.1% refers to the acceptance rate of the *axes* (the 2-of-3 rule) or the *probes* generated from them. The Appendix (Table 6) details a "Human-anchored validation" with 210 binary verdicts, but the link between the 2-of-3 rule in Section 3.1 and the 87.1% figure in Section 5.2 is ambiguous. The claim that the dataset is "validated" relies on this specific 2-of-3 mechanism, but the reported metric (87.1%) appears to be a plausibility score of the *responses* or *judges* rather than the raw acceptance rate of the axes. This conflation weakens the claim of rigorous dataset construction.

Second, the claim in Section 4.2 regarding the Phase Trajectory Fidelity (PTF) metric is slightly overstated. The text states: "PTF is sensitive to phase-order corruption (-8.8 to -23.0 points)." Table 5 ("PTF Metric Validity") supports the -23.0 figure for the \ours{}-32B-DPO model under the "reverse_response" condition. However, for the DeepSeek-V4-Pro model, the drop for "shuffle_response" is only -1.3 points, and for "reverse_response" it is -6.3 points. The range "-8.8 to -23.0" effectively excludes the DeepSeek model's performance, which shows minimal sensitivity to shuffling. By presenting the range as a general property of the metric without qualifying that it applies primarily to the fine-tuned models or specific perturbation types, the claim misrepresents the metric's behavior across the full experimental scope.

Finally, Section 4.3 claims that "DPO improves trajectory direction; SFT often holds a static voice." While the overall scores in Table 3 support the superiority of DPO, the specific claim about "trajectory direction" is not explicitly broken down in the main text. Table 5 does provide "Align" and "Dir" sub-scores, showing a larger drop in "Dir" for the DPO model under perturbation, which supports the claim. However, the text should explicitly reference these sub-metrics to substantiate the specific claim about "direction" rather than relying on the aggregate "Overall" score, which conflates direction with other fidelity aspects.

These issues are primarily matters of precision in reporting rather than fundamental flaws in the science. The data exists to support the claims, but the textual assertions need to be more tightly bound to the specific conditions (e.g., model type, perturbation type) under which the numbers hold.

Action items.

- **[writing]** Clarify if the 87.1% human plausibility in Section 5.2 refers to the 2-of-3 axis acceptance rule in Section 3.1 or a separate probe validation step to avoid ambiguity.
- **[writing]** The claim that PTF sensitivity ranges from -8.8 to -23.0 overgeneralizes; Table 5 shows DeepSeek-V4-Pro has only -1.3 sensitivity to shuffling, contradicting the implied universal range.

- **[writing]** Explicitly cite the 'Dir' sub-metric in Table 5 to support the specific claim that DPO improves 'trajectory direction' rather than just overall fidelity.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively illustrates the character arc concept with clear visual separation between the two time points. However, the caption contains a grammatical error, and the specific chapter citations provided in the figure are factually incorrect regarding the source material.

Figure 2

Figure 2 is a clear and well-structured pipeline diagram that effectively visualizes the two main processes described in the caption: character arc construction and probe generation. The flow is logical, the text is legible, and the use of color and icons aids in distinguishing between different stages and data types without clutter.

Figure 3

The bar chart is visually clear, but the caption is incomplete and fails to define the specific model categories on the x-axis or the units of the y-axis metric.

Figure 4

Figure 4 presents bar charts with unclear Y-axis metrics ('AvgPP') and X-axis groupings that do not align well with the caption's description of an ablation study. The legend terms are undefined in the caption, making it difficult to interpret the source-of-effect analysis.

Figure 5

Figure 5 is a clear screenshot of the human annotation interface described in the caption. It effectively displays the 'agency-communion' axis, including the trajectory phases, evidence summary, and the specific rating questions used for data collection.

Figure 6

The figure displays a hierarchical taxonomy of character arcs but fails to visually support the caption's claim that these axes are clustered and grounded against established scholarship. There is no visual evidence of the clustering or grounding process described.

Figure 7

The figure presents per-model lift data but the caption misleadingly refers to it as 'Overall lift', and the x-axis labels are rotated to the point of being difficult to read.

Figure 8

The figure is a clear screenshot of an annotation interface, but the caption is misleading as it labels the page for 'LLm judges' while the UI elements shown are designed for human annotators.
Action items.

- [**writing**] Figure 1: The caption contains a grammatical error and missing reference ('An example from :') where the dataset or section name should be.
- [**writing**] Figure 1: The timeline labels 'Book 1, Chap 10' and 'Book 5, Chap 130' are factually inconsistent with the Harry Potter series (Book 5 has only 38 chapters).
- [**writing**] Figure 3: The caption contains incomplete text ('Tables and .') and does not define the 'Per-category' groupings (e.g., HER-32B, CoS-ER-8B) shown on the x-axis.
- [**science**] Figure 3: The y-axis label 'Arc lift over best non-Arc (pts)' is ambiguous; it is unclear if 'pts' refers to percentage points, raw score points, or a normalized metric, as no unit definition is provided in the caption.
- [**science**] Figure 4: The caption describes an 'Arc source-of-effect ablation' but the Y-axis is labeled 'AvgPP' (Average Perplexity) without defining the metric or explaining how it relates to the ablation effect.
- [**science**] Figure 4: The X-axis labels 'DS-V4-Flash', 'Qwen3-32B', and 'ArcANE-32B-DPO' do not match the caption's claim of 'three models' if 'DS' is DeepSeek and 'ArcANE' is the proposed method; the specific ablation conditions (e.g., 'Vanilla' vs 'MixedArc') are shown in the legend but the grouping logic on the X-axis is unclear.
- [**writing**] Figure 4: The legend uses 'Vanilla', 'MixedArc', 'ArcHint', and 'Arc', but the caption does not define these terms or explain what 'Arc source-of-effect ablation' entails for each category.
- [**science**] Figure 6: The figure is a sunburst chart showing a taxonomy of character arcs, but the caption claims it shows 'Induced axes are clustered and grounded against established literary or psychological scholarship.' The visual does not display any clustering analysis, statistical grounding, or comparison to external scholarship, making the figure unsupported by the caption's claim.

- [**writing**] Figure 6: The caption cites 'Values in the Wild huang2025values' but the figure itself contains no visual elements (such as citations, references, or source labels) to demonstrate this grounding.
- [**science**] Figure 7: The caption states 'Arc-over-Vanilla Overall lift', but the y-axis is labeled 'Arc lift over Vanilla (pts)'. While similar, the term 'Overall' in the caption implies an aggregate or mean, yet the bars represent specific models (DS-V4-Flash, Qwen3-8B, etc.) rather than a single 'Overall' metric. The figure shows per-model lift, not an overall lift, creating a mismatch between the caption's claim and the data granularity.
- [**writing**] Figure 7: The x-axis labels are rotated at a steep angle and are partially cut off or difficult to read (e.g., 'DS-V4-Flash', 'Qwen3-32B'), reducing legibility.
- [**writing**] Figure 8: The caption identifies the image as an 'Annotation Page for LLm judges', but the screenshot displays a human annotation interface (radio buttons for 'Reasonable'/'Not reasonable', text input for comments) rather than an LLM judge's output or interface.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and specialized terminology that are not consistently defined for a general audience. In the Introduction, the term "RPLAs" (Role-Playing Language Agents) is used immediately without expansion, creating an immediate barrier for readers unfamiliar with the specific sub-field of agent evaluation. Similarly, in Section 3.1, the distinction between "intrapersonal" and "relational" axes is made without plain-language context, assuming the reader possesses specific psychological or literary theory background.

The Evaluation Protocol (Section 4.2) introduces four distinct metrics (APF, RPF, RAE, PTF) and uses their acronyms extensively in the results tables. While the full names are provided, the text does not explicitly state "We refer to these as APF, RPF..." which is a standard convention to ensure clarity. Furthermore, the training methodology in Section 3.5 utilizes "SFT" and "DPO" without defining them, which are standard in RLHF literature but opaque to a broader NLP or general AI audience.

Finally, the validation checks in Section 3.2 (Q-Voice, Q-PhaseFit, Q-Discrim) are introduced as proper nouns without explaining the "Q" prefix or the specific logic behind the naming convention. Replacing these with descriptive terms (e.g., "Voice Consistency Check") or providing a clear definition list would significantly improve accessibility without sacrificing precision.

Action items.

- [**writing**] The manuscript relies heavily on domain-specific acronyms and specialized terminology that are not consistently defined for a general audience. In the Introduction, the term "RPLAs" (Role-Playing Language Agents) is used immediately without expansion, creating an immediate barrier for readers unfamiliar with the specific sub-field of agent evaluation. Similarly, in Section 3.1, the distinction between "intrapersonal" and "relational" axes is made without plain-language context, assuming the read

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a compelling framework for evaluating temporal character consistency, but several logical gaps exist between the presented data and the broad conclusions drawn.

First, the central claim in the Abstract and Section 4.3 that "conditioning on the Character Arc outperforms all other context strategies" is an overgeneralization not fully supported by Table 1 (e000). For the Qwen3-32B model in the "In-Scenario" category, the RAG mode achieves a higher Action Phase-Fidelity (APF) score (59.4) than the Arc mode (57.4). Similarly, in the "In-world" category for the same model, RAG (49.6 APF) slightly trails Arc (54.1 APF), but the "Overall" score for RAG (47.2) is lower than Arc (50.1). The conclusion should be nuanced to specify that Arc is superior *on average* or specifically in *Out-of-World* scenarios where retrieval fails, rather than implying universal dominance across all metrics.

Second, the causal claim that DPO training improves "trajectory direction" (Section 5.3) relies heavily on the Phase Trajectory Fidelity (PTF) metric. However, the validation data in Appendix E002 reveals a weak correlation between PTF and the average of per-phase metrics (APF, RPF, RAE), with an r-squared value of approximately 0.26. This indicates that PTF captures variance largely independent of per-phase accuracy. The paper does not sufficiently argue why PTF is the valid proxy for "direction" when it explains only a quarter of the variance in the component scores. Without a stronger theoretical or empirical link, the assertion that DPO specifically targets "direction" rather than just general phase fidelity remains logically tenuous.

Finally, the ablation study in Section 5.1 uses the "MixedArc" condition to rule out "structured-context bias." The text states that MixedArc "fails to beat Vanilla," but the specific numerical results for MixedArc are not provided in the main text or the visible tables. To logically support the conclusion that the effect is due to the *content* of the arc rather than the *structure* of the context, the review requires the explicit performance data showing MixedArc is statistically equivalent to or worse than the Vanilla baseline.

Action items.

- [**writing**] The paper presents a compelling framework for evaluating temporal character consistency, but several logical gaps exist between the presented data and the broad conclusions drawn. First, the central claim in the Abstract and Section 4.3 that "conditioning on the Character Arc outperforms all other context strategies" is an overgeneralization not fully supported by Table 1 (e000). For the Qwen3-32B model in the "In-Scenario" category, the

RAG mode achieves a higher Action Phase-Fidelity (APF) sco

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the superiority of the "Arc" context mode and the specific benefits of DPO training, particularly in "Out-of-World" scenarios. However, the evidence provided in the text and tables does not fully support the causal or mechanistic explanations offered in the narrative.

First, in Section 4.3 (Main Results), the authors state: "The advantage is largest on Out-of-World probes (+7.7 for DeepSeek-V4-Pro) where retrieval fails." This phrasing implies that the performance gap exists *because* retrieval fails. However, the paper does not present data quantifying the failure rate of the RAG baseline on these specific Out-of-World probes. While it is logically probable that RAG struggles with non-existent source text, the paper treats this as a measured fact rather than a hypothesis. The claim overreaches by presenting an assumption about the RAG mechanism as an empirical finding. The authors should either provide a retrieval success/failure analysis for these probes or soften the language to "where retrieval is less applicable."

Second, in Section 4.4 (Additional Results) and the Conclusion, the authors claim that DPO training "specifically improves In-World and Out-of-World performance" and that "DPO improves trajectory direction; SFT often holds a static voice." While Table 1 and Table 2 show that the DPO-tuned models have higher scores than SFT models, the lift is observed across *all* categories (In-Scenario, In-World, Out-of-World). The text selectively highlights the Out-of-World gain to support the narrative of "trajectory" improvement, but the data does not isolate "trajectory direction" as the specific variable that improved more in Out-of-World scenarios compared to In-Scenario scenarios. The claim that DPO *specifically* targets these out-of-text scenarios is an over-interpretation of a general performance lift. The authors should clarify that the improvement is general, or provide a specific analysis showing that the *magnitude* of improvement for trajectory direction is significantly higher in Out-of-World scenarios than in In-Scenario ones.

Finally, the claim in the Introduction that prior work "fails to assess behavioral shifts in scenarios beyond the source text" is slightly overstated. While TimeCHARA focuses on point-in-time knowledge, the paper does not explicitly rule out that other benchmarks might have incidental coverage of behavioral shifts, even if not their primary focus. A more precise claim would be that prior work does not *systematically* evaluate these shifts across a defined character arc.

These issues are primarily matters of precise wording and avoiding causal overreach rather than fundamental flaws in the experimental design. The data supports the general conclusion that

Arc is effective, but the specific mechanistic explanations for *why* it works best in Out-of-World scenarios require more rigorous backing or softer phrasing.

Action items.

- **[writing]** The claim that the 'Arc' context mode outperforms all others 'especially on Out-of-World scenarios where retrieval fails' (Sec 4.3) overstates the evidence. The paper does not explicitly demonstrate that RAG fails on these specific probes; it only shows Arc is better. The authors must clarify if RAG failure is a measured phenomenon or an assumption, and avoid implying causality without direct evidence of retrieval breakdown.
- **[writing]** The conclusion that DPO training 'specifically improves In-World and Out-of-World performance' (Sec 4.4) is an over-interpretation of the aggregate lift. The data shows a general lift across all categories. The authors should avoid attributing the gain specifically to 'trajectory direction' in out-of-text scenarios without a targeted ablation showing that SFT models fail specifically on trajectory direction in those specific scenarios, rather than just general fidelity.
- **[writing]** The assertion that 'retrieval fails' on Out-of-World probes (Sec 4.3) is a strong claim that requires validation. The paper defines Out-of-World as 'non-source era' but does not provide data showing that RAG systems actually retrieve irrelevant or zero-shot results for these specific prompts. The authors should either provide evidence of retrieval failure or rephrase the claim to 'where retrieval is less likely to be directly applicable' to avoid overreach.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper addresses a novel and important area in evaluating Role-Playing Language Agents (RPLAs) by focusing on temporal character consistency. However, from a safety and ethics perspective, there are significant gaps regarding the handling of harmful content inherent in the source material and the generation of new scenarios.

First, the **Limitations section** (Section 6, lines 240-242) briefly notes that "Period-bound social attitudes in source texts may be reproduced." This is a critical safety concern. Many of the 17 novels included (e.g., 19th-century literature) contain explicit racism, sexism, and other forms of hate speech or discriminatory attitudes. The paper does not describe a mechanism to filter, flag, or handle these outputs during the evaluation process. If the benchmark rewards an RPLA for accurately mimicking a character's use of slurs or discriminatory logic, it risks normalizing such behavior. The authors must explicitly state how they mitigate the risk of the benchmark generating or validating harmful content.

Second, the **dataset construction pipeline** (Section 3.1) relies heavily on LLMs to extract "psychologically impactful events" and define "Character Arcs." There is no detailed description of human oversight specifically regarding the *interpretation* of sensitive psychological states or traumatic events. If the LLMs used for construction hallucinate or introduce biased interpretations of mental health or trauma, the resulting benchmark could propagate these biases. The authors should clarify the extent of human validation for these specific psychological constructs to ensure they do not reinforce harmful stereotypes.

Finally, the **Out-of-World probes** (Section 3.2) involve transposing characters into eras or settings not present in the source text. While this tests generalization, it introduces a risk of generating scenarios that could be offensive or trivialize historical tragedies (e.g., placing a character from a war novel in a context that minimizes the conflict). The paper should confirm that these generated probes underwent a safety review to ensure they do not violate ethical norms regarding the depiction of sensitive historical or social issues.

Addressing these points is essential to ensure the benchmark does not inadvertently become a tool for generating or validating harmful content.

Action items.

- [**writing**] The Limitations section (Sec. 6) acknowledges that 'period-bound social attitudes' in 19th/early-20th-century texts may be reproduced. However, the paper lacks a concrete mitigation strategy for when RPLAs generate hate speech, slurs, or harmful stereotypes inherent to the source material (e.g., in 'Anna Karenina' or 'The Underdogs'). Explicitly state how the evaluation protocol handles or filters such outputs to prevent the benchmark from inadvertently promoting harmful content.
- [**writing**] The dataset construction relies on LLMs to generate 'psychologically impactful events' and 'character arcs' (Sec. 3.1). There is no mention of human oversight regarding the potential for these LLMs to hallucinate or introduce biased interpretations of sensitive psychological traits or traumatic events. Clarify the human-in-the-loop validation process for these specific psychological constructs to ensure they do not reinforce harmful stereotypes.
- [**writing**] The 'Out-of-World' probes (Sec. 3.2) transpose characters into non-source eras. While creative, this risks generating scenarios where characters are placed in contexts that could be interpreted as trivializing historical tragedies or misrepresenting marginalized groups. The paper should confirm that these probes were screened for potential ethical violations or offensive content before evaluation.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a robust dataset construction pipeline with human validation (2-of-3 majority) for the Character Arcs, which strengthens the ground truth for the evaluation. The sample size of 4,601 probes across 17 novels is substantial for this domain. However, the statistical evidence supporting the novelty of the PTF metric is weak. In Appendix PTF Metric Validity, the authors report a Pearson correlation of r=0.51 between PTF and the average of per-phase metrics (APF, RPF, RAE). An r^2 of ~0.26 implies that PTF shares only a quarter of its variance with the per-phase scores, yet the paper treats PTF as a primary, distinct indicator of "trajectory" without demonstrating that it captures unique variance not explained by the component metrics. A regression analysis or variance decomposition is needed to justify PTF as a standalone metric rather than a redundant composite.

Furthermore, the claim that DPO training specifically enhances "trajectory direction" (Section 5.3) relies on a count of probes where the DPO model outperformed the SFT model (1,198/1,750) and a small-scale perturbation study (N=75 in Table 6). The perturbation results show large drops (e.g., -23.0 points for reversal), but without reported p-values or confidence intervals for these differences, it is difficult to assess the statistical significance of the effect size, particularly given the high variance in LLM generation. The ablation study using "MixedArc" (wrong character's arc) is a strong control, but the deterministic hash-based selection of donor characters requires a brief discussion on whether this method could inadvertently select arcs that are structurally similar to the target, potentially biasing the baseline. Finally, while the judge validation shows high correlation with humans (r=0.96), the Mean Absolute Deviation (MAD) of 4.7 points on a 100-point scale suggests a non-trivial error margin that should be contextualized when interpreting the small performance gaps (e.g., +2.2 to +8.4 points) between context modes.

Action items.

- [**science**] The PTF metric validity relies on a correlation of r=0.51 with per-phase averages (Appendix PTF Metric Validity). This indicates PTF captures only ~26% of the variance explained by simple averaging. The authors must provide a stronger statistical justification (e.g., variance decomposition or regression analysis) for why PTF is a distinct and necessary metric beyond a weighted average of APF/RPF/RAE.
- [**science**] The claim that DPO training specifically improves trajectory direction (Section 5.3) is based on a comparison of 1,198/1,750 probes. The evidence lacks a formal statistical test (e.g., paired t-test or Wilcoxon signed-rank) on the magnitude of the PTF improvement to rule out random variation, especially given the small sample size of the perturbation analysis (N=75 in Table 6).
- [**science**] The 'MixedArc' ablation (Section 5.1) uses a deterministic hash to select donor characters. The authors must clarify if this selection process introduces any systematic bias (e.g., selecting characters with similar arc structures) that could artificially inflate the 'Vanilla' baseline performance, thereby underestimating the true effect size of the correct Arc.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in this paper is largely descriptive, relying on mean scores and point-difference comparisons without sufficient inferential statistics to support the strength of the claims.

First, the primary results in Table 1 and Table 2 present only point estimates (e.g., DeepSeek-V4-Pro Arc score of 62.4 vs. Vanilla 52.4). There are no confidence intervals, standard errors, or standard deviations reported for these means. Given the variability inherent in LLM generation and the relatively small number of probes per condition (e.g., 75 probes for the PTF perturbation study in Appendix B), the absence of uncertainty measures makes it difficult to determine if the observed gaps are statistically significant or potentially due to random variance. The claim that "Arc-grounded context achieves the highest Overall score on every model" requires statistical testing (e.g., paired t-tests or Wilcoxon signed-rank tests) to be robust.

Second, the paper performs a large number of pairwise comparisons (6 models × 6 modes × 4 metrics × 3 scenario types) but does not address the multiple comparisons problem. Highlighting the "best" result in each column without correcting for the family-wise error rate (e.g., via Bonferroni or False Discovery Rate) inflates the risk of Type I errors. The authors should clarify if the reported "lifts" (e.g., +7.7 points) remain significant after such corrections.

Third, the validation of the PTF metric (Appendix B) relies on reporting mean drops in scores after perturbation (e.g., -23.0 for reverse_response). While the magnitude of the drop is impressive, the text does not specify the statistical test used to confirm that these drops are significantly different from zero or from the baseline variance. Similarly, the judge validation section reports high Pearson correlations (r 0.92) between the LLM judge and human scores but omits the p-values and confidence intervals for these correlations, which are essential for establishing the judge's reliability.

Finally, the sample sizes for the ablation studies (e.g., N=75 for PTF perturbation, N=210 for human plausibility) are stated, but the power of these tests is not discussed. For the human plausibility check (87.1% pass rate), a binomial test or confidence interval around the proportion would strengthen the claim of validity. The authors should supplement their descriptive statistics with inferential tests and uncertainty estimates to fully support their conclusions.

Action items.

- [**science**] Report confidence intervals or standard errors for all mean scores in Tables 1 and 2. The current presentation of single-point estimates (e.g., 62.4 vs 57.7) lacks a measure of uncertainty, making it impossible to assess the statistical significance of the reported gains without raw data.
- [**science**] Clarify the statistical test used to validate the 'PTF sensitivity' claims in Section 5.2 and Appendix B. The text reports specific point drops (e.g., -23.0) for perturbed conditions but does not state if these differences were tested against a null hypothesis or if the variance across the 75 probes was sufficient to support the conclusion.
- [**science**] Address the multiple comparisons problem. The study evaluates 6 models across 6 context modes and 4 metrics (APF, RPF, RAE, PTF) across 3 scenario types. The paper

highlights ‘best’ results without mentioning any correction (e.g., Bonferroni, Holm-Bonferroni) for the large number of pairwise comparisons performed.

- [**science**] Provide the sample size (N) and degrees of freedom for the Pearson correlation analyses in Appendix B (Judge Validation). While r=0.96 is reported, the statistical power and significance (p-value) of these correlations are missing, which is critical for validating the LLM judge.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high standard of academic writing, with a clear logical flow from the introduction of the problem to the proposed methodology and results. The prose is generally concise and effectively communicates complex concepts regarding character arcs and narrative evaluation. The structure is well-organized, and the use of tables to summarize benchmark comparisons and results significantly aids readability.

However, there are minor areas where sentence-level clarity and consistency can be improved. In Section 3.1, the description of the “Candidate Generation” stage contains a slightly dense sentence regarding the induction of axes. While the meaning is recoverable, a slight restructuring to explicitly separate the LLM’s role from the theoretical grounding would enhance precision. Similarly, in Section 5.1, the comparison between prompting and trained models regarding the “ArcHint” condition suffers from a minor subject-verb ambiguity that could be resolved with a simple rephrase.

Additionally, the manuscript exhibits minor inconsistencies in terminology capitalization, specifically regarding the “Out-of-World” probe category. Standardizing this term throughout the text (e.g., ensuring it is consistently capitalized or lowercased depending on the style guide) would contribute to a more polished final product. The definitions of the evaluation metrics, particularly PTF in Section 4.2, are accurate but could benefit from a brief illustrative example or a direct pointer to the appendix formula to ensure immediate comprehension for readers less familiar with trajectory metrics.

Overall, the writing quality is strong, and these issues are easily addressable with minor revisions. The paper is well-suited for publication pending these small textual refinements.

Action items.

- [**writing**] In Section 3.1 (Character Arc Construction), the phrase ‘inducing intrapersonal... axes grounded in scholarship’ is slightly ambiguous. Clarify whether the axes are induced by the LLM or the human annotators, and ensure the citation placement clearly links to the theoretical grounding.

- [**writing**] In Section 4.2 (Evaluation Protocol), the definition of PTF (‘assesses alignment, direction, and shape') is abstract. Consider adding a brief parenthetical example or a reference to the specific mathematical formulation in the Appendix to improve immediate clarity for the reader.

- **[writing]** In Section 5.1 (Source-of-effect ablation), the sentence 'ArcHint... recovers most of the gain for prompting but only half for trained models' lacks a clear subject for 'recovers'. Rephrase to 'The ArcHint condition recovers...' to ensure grammatical precision.
- **[writing]** Throughout the text, ensure consistent capitalization of 'Out-of-World' vs 'out-of-world'. The manuscript currently uses both forms (e.g., Abstract vs. Section 4.3). Standardize to one style for professional polish.